ENME331 Exam 2 Solver Guide for AI Models

Nov 10, 2026 · @sam

Ground rules for the solver

Solve every problem using only the Midterm 2 equation sheet below plus algebra, geometry, calculus and unit conversion. All material properties are given in the problem, so never look up or assume a property unless the problem says to.

Exam format: 45 minutes, 3 problems at 25 points each, closed book, equation sheet provided. Typical lineup: one energy equation problem (siphon, pump, turbine, head loss), one control volume momentum problem (vanes, elbows, jets, nozzles), one viscous flow problem (Couette or Poiseuille between plates, bearings, Navier-Stokes simplification).

Notation: use the course symbols: p pressure, ρ density, γ = ρg specific weight, V velocity, Q volume flowrate, $\dot{m}$ mass flowrate, A area, D diameter, z elevation (positive up), h_L head loss, h_s shaft head, μ viscosity, τ shear stress, u and v velocity components in x and y.

Required answer format for every problem:

1. One sentence naming the governing principle and why it applies.
2. A sketch description of the control volume or coordinate system, with sign conventions.
3. A "Given" list, all in consistent units.
4. The equation from the sheet written in full, then each term simplified with a reason ("$V_1 \approx 0$, large tank", "p = 0 gage, free jet").
5. Numbers substituted, result in bold with units and direction.
6. A one-line sanity check.

Units: SI uses ρ_water = 1000 kg/m³, g = 9.81 m/s². US uses ρ = 1.94 slug/ft³, γ = 62.4 lb/ft³, g = 32.2 ft/s². Convert cm and mm to m, and in to ft, before substituting.

Midterm 2 equation sheet (the only equations allowed)

Copied from the official sheet attached to the practice exam.

Mass conservation (control volume)

$$\frac{\partial}{\partial t}\int_{cv}\rho\, dV + \int_{cs}\rho(\vec V\cdot\hat n)\, dA = 0$$

Momentum conservation (control volume)

$$\frac{\partial}{\partial t}\int_{cv}\rho\vec V\, dV + \int_{cs}\rho\vec V(\vec V\cdot\hat n)\, dA = -\int_{cs}p\hat n\, dA + \int_{cv}\rho\vec g\, dV + \vec F_{visc}$$

Energy conservation

$$\left[\frac{p}{\rho}+\frac{V^2}{2}+gz\right]_{in} + w_{net\,in} = \left[\frac{p}{\rho}+\frac{V^2}{2}+gz\right]_{out} + \text{loss}$$

$$\left[\frac{p}{\gamma}+\frac{V^2}{2g}+z\right]_{in} + h_s = \left[\frac{p}{\gamma}+\frac{V^2}{2g}+z\right]_{out} + h_L$$

Momentum for a moving control volume (steady, incompressible)

$$\int_{cs}\vec W\rho\vec W\cdot\hat n\, dA = \sum\vec F \qquad \vec W = \vec V - \vec V_c$$

Continuity (differential)

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z} = -\frac{1}{\rho}\frac{D\rho}{Dt}$$

Incompressible: the right side is 0.

Navier-Stokes, incompressible, 2D, steady

$$\rho\left(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x}+\rho g_x+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)$$

$$\rho\left(u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}\right) = -\frac{\partial p}{\partial y}+\rho g_y+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)$$

Shear stress

$$\tau_{yx} = \tau_{xy} = \mu\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)$$

Flow between horizontal parallel plates, u = u(y), v = 0

$$-\frac{\partial p}{\partial x}+\rho g_x+\mu\frac{d^2u}{dy^2} = 0 \qquad y_1 \le y \le y_2$$

$$\frac{\partial p}{\partial x} = \text{constant} \qquad \frac{\partial p}{\partial y} = \rho g_y$$

The sheet notes: specific velocity profiles depend on boundary conditions; either the boundary conditions or the profiles will be provided. All material properties will be provided.

Not on the sheet, but derivable in two lines from it (show the derivation on the exam):

- Couette, no pressure gradient: u = Uy/b, τ = μU/b, q = Ub/2
- Plane Poiseuille, fixed plates, gap 2h: u = (1/2μ)(∂p/∂x)(y² − h²), q = (2h³/3μ)(Δp/ℓ), u_max = 1.5V
- Combined Couette plus Poiseuille, gap b: u = Uy/b + (1/2μ)(∂p/∂x)(y² − by)
- Normal stresses: σ_xx = −p + 2μ(∂u/∂x), σ_yy = −p + 2μ(∂v/∂y)

Control volume methods

Most Exam 2 points come from drawing the right control volume and simplifying the sheet's integral equations term by term.

Continuity (steady, incompressible)

- Q_in = Q_out, so $A_1V_1 = A_2V_2$ for one inlet and one outlet.
- Use the component of V normal to the control surface: $\dot{m} = \rho A|V\cdot n|$. If a CV face is vertical and the flow crosses it at angle θ from horizontal, the flux uses V cosθ.
- Non-uniform profiles: Q = ∫u dA. For a channel of width b: Q = b∫u dy.
- Rising or falling tank: Q_in − Q_out = A_surface(dh/dt).

Linear momentum (steady)

1. Draw the CV around the fluid, cutting inlets and outlets perpendicular to the flow when possible.

2. Pick +x and +y. Write $\Sigma F = \Sigma(\dot{m}V)_{out} - \Sigma(\dot{m}V)_{in}$, one component at a time, keeping signs on velocity components.
3. Forces on the CV: pressure at each cut (use gage pressure; p acts INTO the CV, so p_1A_1 pushes in the inlet flow direction and p_2A_2 pushes against the outlet flow), weight if vertical, and the unknown anchoring or vane force R.
4. Free jets have p = 0 gage at the exit.
5. Solve for R (force ON the fluid). The fluid's force on the device is −R. State direction in words.
6. Per unit width problems (sluice gate): hydrostatic forces $F = \gamma h^2/2$ on each vertical cut.
7. Frictionless flow along a flat plate: the plate can only push normal to itself, so momentum along the plate is conserved. Use it to split the flow ($f_2 - f_3 = \cos\theta$, $f_2 + f_3 = 1$).
8. Moving vane or plate at speed U: use relative velocity W = V − U, and the mass rate hitting it is $\rho A(V - U)$.

Energy equation (head form)

- Pick point 1 and point 2 on a streamline where most terms are known: free surfaces (p = 0, V ≈ 0) and free jets (p = 0).
- Pump: $h_s = +h_p$. Turbine: $h_s = -h_T$. Power: $\dot{W} = \gamma Q h_s$.
- Head loss h_L is always ⩾ 0 and goes on the outlet side.
- "Loss per unit mass" (ft·lbf/slug or J/kg) uses the ρ form: loss = $g \cdot h_L$.
- Siphon or drain: $z_{in} - z_{out} = (1 + K_L)V^2/2g$ when both ends are atmospheric and the tank is large.
- Head loss from a momentum problem: find p_1 from momentum first, then plug into energy.

Pump curve plus system curve: write h_p from energy as a function of Q ($h_p = \Delta z + K_L V^2/2g$ with V = Q/A), write the pump curve as a line, set equal, solve the quadratic, keep the positive root.

Differential analysis and viscous flow

Continuity checks: for a given u and v, confirm $\partial u/\partial x + \partial v/\partial y = 0$. To find a missing component, integrate: $\partial v/\partial y = -\partial u/\partial x$.

Stresses at a point (Newtonian fluid)

1. Take all four derivatives ∂u/∂x, ∂u/∂y, ∂v/∂x, ∂v/∂y and evaluate at the point.
2. $\tau_{xy} = \mu(\partial u/\partial y + \partial v/\partial x)$. Any τ involving z is 0 in a 2D flow.
3. $\sigma_{xx} = -p + 2\mu(\partial u/\partial x)$, $\sigma_{yy} = -p + 2\mu(\partial v/\partial y)$.
4. Sketch: negative normal stresses point into the element; positive τ_{xy} points up on the right face, right on the top face, down on the left, left on the bottom.

Simplifying Navier-Stokes (the standard recipe)

State each assumption and cross out the matching term:

- Steady: no ∂/∂t.
- Fully developed or u = u(y) only: ∂u/∂x = 0, so the whole left side (acceleration) is 0 when v = 0.
- v = 0 everywhere (parallel flow).
- Horizontal plates: $g_x = 0$.
- No pressure gradient (Couette): ∂p/∂x = 0.

What remains: $\mu\, d^2u/dy^2 = \partial p/\partial x$. Integrate twice: $u = (1/2\mu)(\partial p/\partial x)y^2 + C_1 y + C_2$. With no pressure gradient, $u = C_1 y + C_2$.

Boundary conditions: no slip. Fluid at a wall moves with the wall: u = 0 on a fixed wall, u = U on a moving one. Between two fluids: velocity matches and shear stress μ(du/dy) matches at the interface.

Standard results

Situation	Velocity	Key result
Couette, gap b, top plate U	$u = Uy/b$	$\tau = \mu U/b$, $q = Ub/2$
Plane Poiseuille, gap 2h, both fixed	$u = (1/2\mu)(\partial p/\partial x)(y^2 - h^2)$	$q = 2h^3\Delta p/(3\mu\ell)$, $u_{max} = 1.5V$, $\tau_{wall} = h\Delta p/\ell$
Couette plus pressure gradient, gap b	$u = Uy/b + (1/2\mu)(\partial p/\partial x)(y^2 - by)$	u_{max} where du/dy = 0: $y = b/2 - \mu U/[b(\partial p/\partial x)]$
Two fluids, each gap h, top plate U	linear in each layer	$u_{interface} = \mu_1 U/(\mu_1 + \mu_2)$, μ_1 is the top fluid
Pipe (Hagen-Poiseuille), radius R	parabolic	$\Delta p = 8\mu\ell V/R^2$, $\tau_w = 4\mu V/R = (R/2)(\Delta p/\ell)$

Bearings and rotating shafts: if the gap h is much smaller than R, unwrap it into Couette flow. Surface speed U = Rω (ω in rad/s: rev/min × 2π/60). τ = μRω/h. Force = τ(2πRL). Torque = force × R.

Manometer-driven pressure gradient: Δp = (γ_gage − γ_fluid)h_reading over the tap spacing. The side where the gage fluid is pushed down has higher pressure, which sets the sign of ∂p/∂x.

Practice exam: worked solutions

The practice midterm has one problem of each type. These are the templates to follow.

Problem 1: Siphon with head loss (energy equation)

Water siphon, D = 3 cm, $h_L = 0.8V^2/2g$. The top of the siphon is 4 m above the tank surface and 10 m above the outlet, so the outlet is 10 − 4 = **6 m below the tank surface**. (The lower 4 m dimension is the inlet depth and does not enter the answer.)

1. Point 1: tank surface (p = 0, V ≈ 0, z_1 = 6 m). Point 2: outlet jet (p = 0, $V_2 = V$, $z_2 = 0$).
2. Energy: $z_1 = V^2/2g + 0.8V^2/2g$, so $6 = 1.8V^2/2g$.
3. $V = \sqrt{(2 \times 9.81 \times 6/1.8)} = 8.09$ m/s.
4. $Q = (\pi/4)(0.03)^2(8.09) =$ **5.72×10^{-3} m³/s**.

Problem 2: Flow between fixed vanes (momentum)

Q = 1.2 m³/s, p_1 = 15 kPa, p_2 = 10 kPa, V_1 = 10 m/s at 45°, exit at 30°, $A_1 = A_2$. The CV faces are vertical, so the mass flux uses the horizontal velocity components.

1. Continuity across vertical faces: $V_1\cos45° A = V_2\cos30° A$, so $V_2 = 10(0.7071)/0.8660 = 8.165$ m/s.
2. Face area: $A = Q/(V_1\cos45°) = 1.2/7.071 = 0.1697$ m². $\dot{m} = 1200$ kg/s.
3. x-velocities: $V_{1x} = 7.071$ m/s, $V_{2x} = 8.165(0.8660) = 7.071$ m/s. They are equal, so the x-momentum change is zero.
4. x-momentum: $R_x + p_1A - p_2A = \dot{m}(V_{2x} - V_{1x}) = 0$.
5. $R_x = (p_2 - p_1)A = (10{,}000 - 15{,}000)(0.1697) = -849$ N (vanes on water, pointing −x).
6. Water on vanes: **$K_x = +849$ N (in +x)**. Gravity has no x component.

If the CV faces were instead taken perpendicular to the flow (A as the true flow area), the method is the same but A = Q/V and the pressure forces get cosθ factors. Always say which CV you used.

Problem 3: Oil bearing (Couette flow)

Shaft radius R rotating at ω inside a fixed casing, gap h, $u = C_1 y + C_2$ with y measured up from the fixed casing.

(a) Boundary conditions: u(0) = 0 (fixed casing) and u(h) = Rω (no slip at the shaft surface).

- u(0) = 0 gives $\mathbf{C_2 = 0}$.
- u(h) = Rω gives $\mathbf{C_1 = R\omega/h}$.

(b) $\tau = \mu \, du/dy = \mu R\omega/h = 500$ N/m².

$\omega = \tau h/(\mu R) = 500(0.001)/(0.4 \times 0.2) =$ **6.25 rad/s** (about 59.7 rev/min).

Common traps

These caused wrong or near-miss answers on recent homework. Check each before finalizing.

Trap	Rule
Using half-gap vs full gap	Plane Poiseuille formulas use h = half the gap. A 4 mm gap means h = 2 mm.
Elevation from a figure	Subtract dimensions carefully: in the siphon, the outlet is 10 − 4 = 6 m below the surface, not 10 m.
Force on fluid vs force on device	Momentum gives R on the fluid. The answer asked for is often the opposite sign.
Pressure force direction	Gage pressure always pushes into the CV at each cut. Free jets have p = 0.
Velocity component in mass flux	Use V·n. A vertical CV face crossed at angle θ uses V cosθ.
Jet angle reference	Check whether an angle is from the horizontal, the vertical or the axis. Try the other reading if an answer is rejected.
rev/min to rad/s	ω = rpm × 2π/60. Forgetting this is a factor of about 9.5.

Trap	Rule
Energy loss units	Per unit mass (ft·lbf/slug, J/kg) is g × head loss in length units.
Moving plate	Use relative speed V − U for both the momentum flux and the mass rate hitting the plate.
Two-fluid Couette	Match velocity and shear stress at the interface; μ_1 is whichever fluid touches the moving plate.
Sign of dp/dx from manometer	Gage fluid pushed down marks the high pressure side.
Flowrate vs velocity	Integrate across the full gap; y disappears from q.

When a figure is ambiguous, state the reading used and give the alternate answer in one line.

Worked example bank

Solved homework problems from this unit. Use them to calibrate method and magnitude.

#	Problem	Key equations	Answer
1	Channel: uniform V over 0.75 ft becomes $u = 4y - 2y^2$ over h = 0.92 ft	$0.75V = \int_0^h u\,dy$	V ≈ 1.56 ft/s
2	Lake Mead: 30,600 cfs in, 8,000 out, 225 mi²	Q_net = A dh/dt	0.311 ft/day
3	Horizontal 180° elbow and nozzle, D 300 to 160 mm, p_1 = 100 kPa, V_1 = 3 m/s	$p_1A_1 + K_x = \dot{m}(-V_2 - V_1)$	K_x ≈ 9940 N to the left
4	Jet engine thrust stand, 676 to 1640 ft/s, 10 ft², 11.4 psia, 472°R	$\rho = p/RT$; $F + p_1A_1 = \dot{m}(V_2 - V_1)$	F ≈ 17,960 lb
5	Air jet, 7 in, on a cone (60°), F_A = 5 lb	$F = \rho AV^2(1 - \cos 60°)$	Q ≈ 33.5 ft³/s

#	Problem	Key equations	Answer
6	Wind tunnel wake, 30 m/s with linear 20 to 30 m/s wake over ±1 m	x-momentum with side outflow	D = 280 N downstream
7	180° elbow and nozzle, 12 to 6 in, 15 psi, 5 ft/s	loss = $p_1/\rho + V_1^2/2 - V_2^2/2$	926 and 200 ft·lbf/slug
8	Filtered end cap, 0.12 to 0.10 ft², 10 ft/s, jet 30° below horizontal, R_y = 60 lb	y-momentum for p_1, then energy	h_L ≈ 2.35 ft
9	10 mm water jet tipping an 8 N block, jet 45 mm up, block 15 mm thick	$\rho AV^2(0.045) = 8(0.0075)$	$Q \approx 3.24 \times 10^{-4}$ m³/s
10	Air jet 33 m/s, 30 mm, on a 30° plate; plate moving at 11 m/s	$F = \rho AV^2\sin30°$; relative speed	0.473 N; $f_2 = 0.933$, $f_3 = 0.067$; 0.210 N
11	Turbine, 40 m head, 1 m exit at 6 m/s	$W' = \gamma Q(40 - V^2/2g)$	1.76 MW
12	Pump curve $h_p = 200 - 100Q$, 100 ft downhill, K_L = 23, 4 in pipe	$200 - 100Q = -100 + 46.9Q^2$	$Q \approx 1.68$ ft³/s
13	Sluice gate, 11 ft to 1.5 ft, 4 ft/s; then closed	$F_1 - F_2 - R = \rho q(V_2 - V_1)$	1543 lb/ft; 3775 lb/ft
14	$u = 12xy^2 - 6x^3$, $v = 18x^2y - 4y^3$, glycerin, p = 6 kPa at (0.5, 1)	$\sigma = -p + 2\mu\partial u/\partial x$; $\tau = \mu(\partial u/\partial y + \partial v/\partial x)$	$\sigma_{xx} = -5.98$ kPa, $\sigma_{yy} = -6.02$ kPa, $\tau_{xy} = 45$ Pa
15	$V = (3xy^2 - 4x^3)i + (12x^2y - y^3)j$ at (1, 2), μ = 0.013	$\tau_{xy} = \mu(6xy + 24xy)$	$\tau_{xy} = 0.78$ N/m²; $\tau_{yz} = \tau_{zx} = 0$
16	$u = x^2$ along the x-axis streamline	continuity; $a = u\partial u/\partial x$; x Navier-Stokes	$\partial v/\partial y = -2x$; $a = 2x^3\,i$; $\partial p/\partial x = 2\mu - 2\rho x^3$
17	$u = ay - b(cy - y^2)$, v = w = 0	x Navier-Stokes; τ(0) = 0	$\partial p/\partial x = 2\mu b$; a = bc

#	Problem	Key equations	Answer
18	SAE 30 oil between fixed plates, 4 mm gap, 31 kPa/m	$q = 2h^3\Delta p/(3\mu\ell)$ with $h = 2$ mm	$q \approx 4.35 \times 10^{-4}$ m²/s; $\tau = 62$ N/m² in the flow direction; $u_{max} \approx 0.163$ m/s
19	Fixed plates 0.7 in apart, $V = 0.4$ ft/s, $\mu = 0.008$	$\Delta p/\ell = 3\mu V/h^2$	11.3 lb/ft³; $u_{max} = 0.6$ ft/s
20	Simple Couette, gap b, top plate U	$\mu\, d^2u/dy^2 = 0$	$u = Uy/b$; $q = Ub/2$
21	Two fluids, equal layers h, top plate U	match u and τ at interface	$u_i = \mu_1 U/(\mu_1 + \mu_2)$
22	Couette plus pressure gradient, 1 in gap, $U = 0.03$ ft/s, manometer 0.1 in over 6 in	$du/dy = 0$	$y_m \approx 0.889$ in
23	Vertical shaft, 75 mm, 0.25 mm gap, 160 mm long, 30 rev/min, $\mu = 0.2$	$\tau = \mu R\omega/h$; $T = \tau(2\pi RL)R$	$T \approx 0.133$ N·m
24	4 mm tube, 5 m fully developed, $V = 1$ m/s, $\mu = 0.015$, $\rho = 1200$	$\Delta p = 8\mu\ell V/R^2$; $\tau_w = 4\mu V/R$	150 kPa; 30 N/m²